

12_open_source_perspective

Genesis_Medicine: an open-source AI pipeline for Korean traditional medicine drug discovery — a 50+ tool integrated stack and design philosophy

HanCheongWoo ^{1,2,3}

¹ Genesis_Medicine Lab, Seoul, Republic of Korea ² HAN PREDICT, Inc.; <https://hanpredict.com> ³ Recover Korean Medicine Clinic; <https://recover-clinic.kr>

Code: https://github.com/crazat/genesis_medicine · Correspondence: admin@hanpredict.com

Manuscript type: Resource / perspective; **Target preprint:** ChemRxiv (primary);
License: CC-BY 4.0 **Status:** System / pipeline description; underlying experimental validation reported elsewhere

Abstract

Modern AI-driven drug discovery is increasingly built on a heterogeneous stack of open-source tools — generative chemistry (REINVENT, fragment-based), property prediction (ADMET-AI, Chemprop), structure prediction (AlphaFold-3, Boltz-2, Protenix), molecular dynamics (OpenMM, openmmtools), and absolute / relative binding free energy estimation. Specialized application domains require tool selection, integration, and adaptation specific to that domain's data and constraints. We describe **Genesis_Medicine**, an integrated open-source pipeline for **AI-driven Korean traditional medicine (한약·생약) drug discovery**, comprising approximately 50+ tools / modules organized into **12 functional tiers** and exposed through a unified natural-language agent interface. The pipeline supports the complete workflow from natural-product compound curation, generative scaffold-hopping, ADMET filtering, and structure-based virtual screening, through molecular-dynamics validation and corrected absolute binding free energy estimation, to TRIPOD-AI-compliant manuscript reporting. We discuss the design philosophy — including 3-pillar institutional integration with HAN PREDICT, Inc. (AI healthcare platform) and Recover Korean Medicine Clinic; commitment to honest in silico reporting; explicit acknowledgment of method limitations including MMP-1 zinc handling and Boltz-2 binary-classifier vs IC₅₀ distinction. The pipeline is open-source under Apache-2.0 at https://github.com/crazat/genesis_medicine.

Keywords: open-source pipeline, Korean traditional medicine, AI drug discovery, REINVENT4, Boltz-2, ADMET-AI, ABFE, OpenMM, natural products, integrated stack.

Plain-language summary

Modern AI-driven drug discovery uses many specialized software tools, mostly free and open-source. Combining these tools into a working pipeline for a specific application — in our case, AI-driven Korean traditional herbal medicine drug discovery — requires careful selection and integration. We describe an open-source pipeline (Genesis_Medicine) that brings together about 50 tools to support the complete workflow from herb-derived compound to in silico drug-candidate report. The code is freely available for academic and commercial use. **This paper is a description of the pipeline; no clinical or laboratory measurements are reported here.**

1. Introduction

1.1 The fragmented AI drug-discovery stack

Modern AI drug discovery requires composition of multiple specialized tools, each excellent at one task and not designed for end-to-end integration. The recent open-source landscape is rich: Boltz-2, AlphaFold-3, Protenix for structure prediction; REINVENT-4, SATURN, f-RAG for generative chemistry; ADMET-AI, Chemprop 2 for properties; OpenMM 8 + openmmtools for MD and free energy; PoseBusters, GNINA for pose validation; Open Targets and DGIdb for disease-target context. Each tool has its own input format, environment requirements, and output structure. End-to-end use requires careful glue-code, validation, and domain-specific configuration.

1.2 The Korean traditional-medicine domain

The Korean traditional-medicine (한약·생약) domain has specific requirements that are not fully addressed by general-purpose pipelines:

- **Natural-product chemistry** (saponins, terpenoids, anthraquinones, polyphenols, alkaloids) — atypical scaffolds for which Caucasian-cohort-trained models may underperform.
- **Korean Pharmacopoeia (KP) and Korean Herbal Pharmacopoeia (KHP)** — specific listings, quality standards, traditional indications.
- **다중성분 복합처방 (multi-component formulations)** — Korean medicine prescribes combinations, not single compounds; pipeline must support combination scoring.
- **체질의학 / 사상의학 (constitutional medicine)** — patient stratification by traditional constitutional theory.
- **Regulatory: MFDS, 의료법 §56, 화장품법** — Korean-specific compliance.

1.3 Genesis_Medicine: design goals

We assembled the Genesis_Medicine pipeline with the following goals:

1. **End-to-end integration** of generative → property → structure → MD → ABFE → manuscript.
 2. **Korean traditional-medicine domain orientation:** KP / KHP herb-compound mapping, 한약 처방 network analysis, Korean PGx panel.
 3. **Honest in silico reporting:** TRIPOD-AI compliance, explicit limitations, retraction of incorrect prior results when discovered.
 4. **Open-source under permissive license:** Apache-2.0; commercial-use compatible; avoids GPL viral.
 5. **Natural-language interface:** a single LLM-agent that accepts natural-language requests and dispatches to the appropriate pipeline tier.
 6. **3-pillar institutional integration:** with HAN PREDICT (AI healthcare platform) and Recover Korean Medicine Clinic; clinical feedback loop closed.
-

2. Pipeline architecture: 12 functional tiers

The Genesis_Medicine codebase organizes approximately 50+ tools / modules into 12 functional tiers. Each tier is exposed as a natural-language tool callable through a ChemCrow-style agent wrapper [1].

Tier 0 — Foundation (~7 modules)

Open-source SOTA stack: Boltz-2 cofold, REINVENT 4 generative, ADMET-AI predictions, RDKit chemistry, OpenMM 8 MD, openmmtools alchemical sampling, MACE-OFF24 ML potential. Standard conda / pip / uv installation under Korean GPU stack (CUDA 12.8, RTX 5090 Blackwell).

Tier 1 — Acceleration / Quality (~6 modules)

Protenix-v2, OpenFold-3 ensemble (Boltz-2 consensus); AlphaFlow-Lit; CarsiDock-Cov for covalent-warhead screening; Boltz-ABFE direct path; AIMNet2 ML potential; DeepRetro retrosynthesis.

Tier 2 — Skin-specific Adapters (~6 modules)

PanDerm (skin lesion vision-language; integrated as skin-classification adapter); ChemCrow LLM agent wrapper; stratum-corneum MD setup; Bayesian SAR optimization; ESM3 protein function; scFoundation single-cell.

Tier 3 — Multi-target / Cryptic-site (~5 modules)

Four added skin-fibrosis targets (LPA1 / $\alpha\text{v}\beta 6$ / YAP1 / TAZ); DGAT herb-DDI module; QM/MM ABFE book-ending correction; spatial-transcriptomics adapter; mechanotransduction multi-tier analysis.

Tier 4 — Kinetics / SAR loops (~5 modules)

k_off residence-time MD setup; Chou-Talalay synergy + topical PBPK; self-improvement agent loop; Enamine REAL Tanimoto matching; Lunit / VUNO collaboration analysis.

Tier 5 — Causal / Stratified evidence (~5 modules)

Causal inference (DoWhy / EconML); papillary-reticular fibroblast stratification; skin microbiome AI; KEDD foundation-model design; federated-learning Korean-medicine-network architecture.

Tier 6 — Korean PGx + skin niche (~5 modules)

Korean PGx panel; mitochondrial PGC-1 α / TFAM; CellChat fibroblast-niche LR pairs; smartphone scar assessment; OliX siRNA collaboration.

Tier 7 — Partnership scaffolds (~5 modules)

ExoCoBio collaboration scaffold; EZH2 epigenetic adapter; NLRP3 inflammasome; Pro-Hyp triple combination; PBM LED Recover protocol.

Tier 8 — Reporting + Health Economics (~5 modules)

TRIPOD-AI checklist agent tool; SHAP / Grad-CAM explainability; 동의보감 herb miner; D+Q senolytic hair-follicle; HIRA QALY framework.

Tier 9 — Diagnostic + Specialized (~5 modules)

facial_dx integration bridge; PROTAC EMB-3 TGFB1 degrader designer; chronotherapy 자오류주; OCT bridge to facial_dx; engineered C. acnes probiotic; ESG / green-chemistry analyzer.

Tier 10 — Safety + MLOps + Knowledge (~5 modules)

Hallucination guard (jailbreak + medical fact citation); MLOps model registry (FDA PCCP); knowledge-graph completion; PA-EWC continual learning; multi-omics fusion.

Tier 11 — Frontier / Strategic (~6 modules)

NaFM natural-product foundation model; Bayesian adaptive trial design + n-of-1 + MFDS DTx; KIPRIS + USPTO patent prior-art; SkinAge clock; Chai-1 ensemble; JUMP-CP morphology; MFDS K-bio M&A landscape.

Total modules (Tier 0 through Tier 11): approximately 50, with substantial cross-tier dependencies. The natural-language agent interface (`src/genesis_medicine/agents/`) exposes all of these as 70 callable tools.

3. Design philosophy

3.1 Honest in silico reporting

A central commitment is **honesty about what is and is not validated**. Specific mechanisms:

- All preprints carry a “**in silico, wet-lab pending**” flag in the abstract.
- **TRIPOD-AI checklist** is run on each preprint as a self-audit.
- **Earlier incorrect results are explicitly retracted** when discovered (e.g., the earlier ABFE values on EMB-3 / Embelin under a complex-leg-only protocol [2]).
- **MMP-1 zinc handling caveat** is consistently flagged across all preprints involving MMP-1 [2,3].
- **Boltz-2 binary-classifier vs IC₅₀** distinction is consistently made.
- **자운고 narrative** correction (chemistry-based clarification of *Embelia ribes* / 자단 distinction from 자운고 / 자초) [4].

3.2 Apache-2.0 open-source

The pipeline code is open-source under Apache-2.0, which is permissive (commercial-use compatible, no GPL viral, no requirement to open-source derivative works). This contrasts with some traditional-medicine database licenses (HERB CC-BY-NC, TCMSP non-commercial) that we explicitly exclude from commercial-build profiles. The license model supports both academic publication and Recover Korean Medicine Clinic’s commercial use.

3.3 3-pillar institutional integration

The pipeline is operated across three affiliated entities: - **HAN PREDICT, Inc.** (hanpredict.com) — AI healthcare platform; provides Smart Charts AI EHR + 3D facial diagnostics infrastructure. - **Genesis_Medicine Lab** — R&D division; develops and maintains the pipeline. - **Recover Korean Medicine Clinic** — clinical practice (강남, 2026-08-15 opening); provides clinical-context feedback loop.

This 3-entity integration is described in detail in our companion workflow preprint [5].

3.4 Korean-specific orientation

The pipeline is oriented to the Korean clinical, regulatory, and traditional-medicine context: - **Regulatory**: PIPA + Korean AI Basic Act + 의료법 + 화장품법 + MFDS pathways. - **Pharmacopoeial**: KP (Korean Pharmacopoeia) + KHP (Korean Herbal Pharmacopoeia) + Donguibogam digitization. - **Linguistic**: Korean clinical-context comments and natural-language tool descriptions. - **PGx**: Korean-population variant frequencies in pharmacogenomic panel.

This orientation does not preclude international application — the pipeline modules are individually general-purpose — but it does motivate specific configurations and weight choices.

4. Scaling and reproducibility

4.1 Hardware

The pipeline runs on a single-GPU configuration (1 × NVIDIA RTX 5090, 32 GB, CUDA 12.8). For reference, a typical end-to-end EMB-3-style scaffold-hopping case study (REINVENT4 100 samples + ADMET filter + Boltz-2 cofold 30 jobs + 10 ns MD + corrected ABFE 64 ns) runs in approximately 24-30 hours on this hardware.

4.2 Reproducibility

- **Conda environments** locked: `genesis-md` (MD-specific) and main `.venv` (Python 3.11).
- **Configuration as code**: `conf/*.yaml` covers all major pipeline parameters.
- **Random seeds** documented per-run.
- **Result archival**: per-run JSON outputs include configuration, version metadata, wall-clock, and key results.

4.3 License compatibility for commercial use

Two build profiles: - **Research** profile: includes HERB 2.0, TCMSP, BATMAN-TCM (non-commercial) for network-pharmacology research. - **Commercial** profile: excludes the above; relies only on permissively-licensed COCONUT 2.0 (CC0), NPASS 3.0 (CC-BY), NPAtlas (CC-BY), Dr. Duke's database (USDA public domain). Boltz-2 (MIT), REINVENT 4 (Apache-2.0), ADMET-AI (MIT), OpenMM 8 (MIT), openmmtools (MIT), RDKit (BSD-3) are all commercial-use-compatible.

This dual-profile approach permits academic and commercial use with appropriate licensing hygiene.

5. Limitations and forward path

1. **No clinical efficacy data** in any of the application studies; all are in silico hypothesis-generation.
2. **MMP-1 zinc handling** is a known limitation; ZAFF integration is planned.
3. **NaFM weights** (natural-product foundation model) are not yet integrated; placeholder fallback to Morgan fingerprint embeddings.
4. **Chai-1 integration** is interface-only; not benchmarked against Boltz-2 + Protenix consensus.
5. **JUMP-CP morphology** integration is offline-mock; AWS S3 data download (~50 GB) is the next forward step.
6. **Patent prior-art search** is offline-heuristic; KIPRIS + USPTO API integration requires API keys.

The Genesis_Medicine pipeline is a living open-source project. Contributions, issue reports, and forks are welcome under Apache-2.0.

6. Conclusions

Genesis_Medicine is an open-source, end-to-end AI pipeline for Korean traditional medicine drug discovery, integrating approximately 50+ tools across 12 functional tiers under a unified natural-language agent interface. The pipeline supports REINVENT4 generative scaffold-hopping, ADMET filtering, Boltz-2 cofold, MD validation, and corrected ABFE estimation, with explicit Korean traditional-medicine domain orientation, honest in silico reporting, and 3-pillar institutional integration with HAN PREDICT, Inc. (AI healthcare platform) and Recover Korean Medicine Clinic. The pipeline is available under Apache-2.0 at https://github.com/crazat/genesis_medicine.

We invite collaboration and contribution from the AI-drug-discovery community.

Acknowledgments / Contributions / Competing interests / Data availability

Same standard text. Code: https://github.com/crazat/genesis_medicine.

Figures

Figure 1. Genesis_Medicine 12-tier pipeline architecture overview. Approximately 50+ modules organized into 12 functional tiers from Tier 0 (Foundation: Boltz-2, REINVENT4, ADMET-AI, OpenMM, openmmtools, MACE-OFF24, RDKit) through Tier 11 (Frontier: NaFM, Bayesian trial design, patent prior-art, SkinAge clock, Chai-1, JUMP-CP, MFDS K-bio landscape). Apache-2.0 open-source.

Genesis_Medicine 12-tier pipeline (50+ modules)
Apache-2.0 open-source · github.com/crazat/genesis_medicine

Tier 0	Foundation Boltz-2, REINVENT4, ADMET-AI, OpenMM, openmmtools, MACE-OFF24, RDKit
Tier 1	Acceleration / Quality Protenix, OpenFold3, AlphaFlow-Lit, CarsiDock-Cov, Boltz-ABFE, DeepRetro, AIMNet2
Tier 2	Skin-specific Adapters PanDerm, ChemCrow agent, Stratum corneum MD, Bayesian SAR, ESM3, scFoundation
Tier 3	Multi-target / Cryptic LPA1/avb6/YAP1/TAZ, DGAT-DDI, QM/MM ABFE, Spatial-T, mechanotransduction
Tier 4	Kinetics / SAR loops k_off MD, Chou-Talalay PBPK, self-improvement loop, Enamine REAL, Lunit/VUNO
Tier 5	Causal / Stratified DoWhy/EconML, papillary-reticular fibroblast, microbiome AI, KEDD, federated-learning
Tier 6	Korean PGx + niche Korean PGx panel, mitochondrial, CellChat, smartphone, Olix siRNA
Tier 7	Partnership scaffolds ExoCoBio, EZH2 epigenetic, NLRP3, Pro-Hyp, PBM LED
Tier 8	Reporting + HE TRIPOD-AI, SHAP/Grad-CAM, 동의보감 miner, D+Q senolytic, HIRA QALY
Tier 9	Diagnostic + Specialized facial_dx integration, PROTAC, chronotherapy, OCT, engineered probiotic, ESG
Tier 10	Safety + MLOps + KG hallucination guard, MLOps registry, KG completion, PA-EWC, multi-omics fusion
Tier 11	Frontier / Strategic NaFM, Bayesian trial + DTx, KIPRIS+USPTO patent, SkinAge, Chai-1, JUMP-CP, MFDS K-bio

Figure 1: 12-tier pipeline

References

[1] M. Bran A, et al. ChemCrow: augmenting large-language models with chemistry tools. *Nat Mach Intell* 2024, 6, 525–535. [2] HanCheongWoo. Calibrated absolute binding free energy pipeline. ChemRxiv preprint, 2026. [3] HanCheongWoo. EMB-3 case study. ChemRxiv preprint, 2026. [4] HanCheongWoo. *Embelia ribes* (Vidanga, 자단) revisited. bioRxiv preprint, 2026. [5] HanCheongWoo. Integrated AI workflow for a Korean medicine clinic. medRxiv preprint, 2026. [6] Wohlwend J, et al. Boltz-2: an open-source biomolecular structure and binding affinity model. Preprint, 2024. [7] Loeffler HH, et al. REINVENT 4: modern AI-driven generative molecule design. *J Cheminform* 2024, 16, 20. [8] Swanson K, et al. ADMET-AI. *Bioinformatics* 2024, 40, btae416. [9] Eastman P, et al. OpenMM 8. *J Chem Theory Comput* 2024, 20, 8226–8235. [10] Chodera JD, et al. openmmtools (v0.26). 2026.

v0.1 draft, 2026-04-26 · ~3,200 words · CC-BY 4.0